Jupyter Organization Tips, Part 2

DATA SCIENCE BOOTCAMP

NASHVILLE SOFTWARE SCHOOL

Use Markdown

Jupyter notebooks are a form of literate programming, which means that they can contain snippets of code interspersed with explanations of how they work in natural language.

These explanations can be formatted using Markdown.

See these two guides:

- Markdown Examples
- Basic Syntax

Explain and Interpret Your Findings

Don't just give numbers as answers or findings.

At minimum, include units, but also include whether you are presenting a proportion or a percentage.

Ratios should have a clear numerator and denominator. Borrowing from this guide, if you're reporting queries per user, do you mean queries/users with a query, queries/users who visited Google today, queries/users with an active account, or something else?

Acknowledge and count your filters; clearly specify what and when you are filtering and indicate the count/fraction of observations removed.

Avoid Hard-Coded Values

You should have a goal of having reusable code, meaning that you don't have to modify it if you apply it to a different subset or if the underlying data changes.

Make use of variables to extract any key values that you are using.

Avoid Hard-Coded Values

Example: Let's say we wanted to find the top 10% of countries in terms of GDP Per Capita.

If the numbers are revised or additional data is added to the dataset, the following code could potentially need to be updated.

```
gdp_df_2021['GDP_Per_Capita'].quantile(0.9)
```

```
49754.04773704288
```

```
gdp_df_2021[gdp_df_2021['GDP_Per_Capita'] >= 49754.04773704288]
```

Avoid Hard-Coded Values

Example: Let's say we wanted to find the top 10% of countries in terms of GDP Per Capita.

This version would not need to be updated even if the data changes.

```
percentile_90 = gdp_df_2021['GDP_Per_Capita'].quantile(0.9)
percentile_90
```

```
49754.04773704288
```

```
gdp_df_2021[gdp_df_2021['GDP_Per_Capita'] >= percentile_90]
```

Make Use of f-strings

f-strings allow you to embed expressions and variables directly into strings.

See this guide.

```
print(f'The 90th percentile of GDP per capita values in 2021 was {percentile_90}.')
```

```
The 90th percentile of GDP per capita values in 2021 was 49754.04773704288.
```

You can even apply formatting. See this guide for more information.

For example, to round to one decimal place:

```
print(f'The 90th percentile of GDP per capita values in 2021 was {percentile_90:0.1f}.')
```

```
The 90th percentile of GDP per capita values in 2021 was 49754.0.
```

Make Use of f-strings

This also works with multiple variables.

```
year = 2021
```

```
print(f'The 90th percentile of GDP per capita values in {year} was {percentile_90:0.1f}.')
```

```
The 90th percentile of GDP per capita values in 2021 was 49754.0.
```

Summary

Checklist for a well-organized and readable notebook:

- All imports are contained at the top of the notebook
- Runs error-free from top to bottom
- Uses markdown and comments
- Contains clear structure and organization; uses headers and subheaders to break up different sections of the notebook
- Unnecessary code cells are removed
- Uses descriptive variable names
- Will, with some exceptions, still run even if the underlying data is updated
- Explains and interprets findings
- Contains properly labeled visualizations (titles, axes, legends, etc.)
- Data provenance - includes information about the data sources
- **Advanced:** Checks assumptions about the dataset and automatically flags issues
- **Advanced:** Modular code - uses functions to avoid code repetition

Assignment

By end of day Sunday, 10/26, push your final notebook to GitHub and DM Michael that you're done.

This final version should be cleaned and organized and ready for feedback.